I. TRANSPORT DUE TO ELECTROSTATIC PERTURBATIONS

For the treatment of electron component, its is sufficient to use the drift-kinetic approximation. Drit kinetic equation using for velocity space variables besides the gyrophase ϕ also the magnetic moment μ and parallel velocity $v_\parallel$ where

$$\mu = \frac{m_\alpha v_\perp^2}{2B} = \frac{e_\alpha J_\perp}{m_\alpha c}$$

and $J_\perp$ is the perpendicular adiabatic invariant, is of the form

$$\frac{\partial f}{\partial t} + \mathbf{v}_g \cdot \nabla f + \dot{v}_\parallel \frac{\partial f}{\partial v_\parallel} = \hat{L}_C f. \tag{1}$$

Here, the guiding center velocity is

$$\mathbf{v}_g = \frac{v_\parallel \mathbf{B}}{B_\parallel^*} + \frac{\mathbf{B} \times \nabla\left(\mu B + e_\alpha \Phi\right)}{m_\alpha \omega_c B_\parallel^*} + \frac{v_\parallel^2}{B_\parallel^*} \nabla \times \frac{\mathbf{B}}{\omega_c}, \qquad B_\parallel^* = B + v_\parallel \mathbf{h} \cdot \nabla \times \frac{\mathbf{B}}{\omega_c}, \tag{2}$$

$\mathbf{h} = \mathbf{B}/B$ and $\omega_c = e_\alpha B/(m_\alpha c)$ are unit vector along the magnetic field and the cyclotron frequency, respectively,

$$\dot{v}_\parallel = -\frac{1}{m_\alpha B_\parallel^*}\left(\mathbf{B} + v_\parallel \nabla \times \frac{\mathbf{B}}{\omega_c}\right) \cdot \nabla\left(\mu B + e_\alpha \Phi\right), \tag{3}$$

and $\hat{L}_C$ is the collision opertor. We ignore the magnetic drift in favour of electric drift so that $B_\parallel^* \to B$ and

$$\mathbf{v}_g \to v_\parallel \mathbf{h} + \frac{c\mathbf{h} \times \nabla \Phi}{B}, \qquad \dot{v}_\parallel \to -\frac{1}{m_\alpha} \mathbf{h} \cdot \nabla\left(\mu B + e_\alpha \Phi\right). \tag{4}$$

A. Quasilinear theory

We assume purely electrostatic perturbations such that magnetic field $\mathbf{B}$ is axisymmetric and electrostatic potential $\Phi = \Phi_0 + \delta\Phi$ contains the unperturbed, axisymmetric potential $\Phi_0 = \Phi_0(r)$ and the perturbation $\delta\Phi$. We stay in a straight cylinder geometry where the unperturbed motion is simple,

$$\mathbf{v}_{g0} = v_\parallel \mathbf{h} + \mathbf{V}_E, \qquad \mathbf{V}_E = \frac{c\Phi_0'(r)}{B} \mathbf{h} \times \nabla r, \qquad \dot{v}_{\parallel 0} = 0. \tag{5}$$

Respectively, the perturbation of the phase space velocity $\delta\mathbf{v}_g = \mathbf{v}_g - \mathbf{v}_{g0}$ is driven by $\delta\Phi$,

$$\delta\mathbf{v}_g = \frac{c\mathbf{h} \times \nabla \delta\Phi}{B}, \qquad \delta\dot{v}_\parallel = -\frac{e_\alpha}{m_\alpha} \mathbf{h} \cdot \nabla \Phi. \tag{6}$$

We use the normalized coordinates (r, ϑ, φ) where $\varphi = z/R_0$ is the normalized cylinder length and $R_0 = const$ is the major radius, so that $\sqrt{g} = rR_0$. In zero order over the perturbation amplitude, we take a local Maxwellian,

$$f_0(r, \mu, v_\parallel) = \frac{n_\alpha(r)}{(2\pi m_\alpha T_\alpha(r))^{3/2}} \exp\left(-\frac{2\mu B(r) + m_\alpha v_\parallel^2}{2T_\alpha(r)}\right), \tag{7}$$

which obviously satisfies the unperturbed steady state equation

$$\mathbf{v}_{g0} \cdot \nabla f_0 + \dot{v}_{\parallel 0} \frac{\partial f_0}{\partial v_\parallel} = \hat{L}_C f_0.$$

We assume the perturbation of the form

$$\delta\Phi(t, r, \vartheta, \varphi) = \mathrm{Re} \sum_{\mathbf{m}} \Phi_{\mathbf{m}}(r) \mathrm{e}^{im\vartheta + in\varphi - i\omega_{\mathbf{m}} t}, \tag{8}$$

where $\mathbf{m} = (m, n)$. We similarly expand the perturbation of the distribution function, $\delta f = f - f_0$, which we treat in the linear order (omitting the terms quadratic in $\delta\Phi$),

$$\delta f(t, r, \vartheta, \varphi, \mu, v_\parallel) = \mathrm{Re} \sum_{\mathbf{m}} f_{\mathbf{m}}(r, \mu, v_\parallel) \mathrm{e}^{im\vartheta + in\varphi - i\omega_{\mathbf{m}} t}, \tag{9}$$

and obtain equations for Fourier amplitudes,

$$(\mathbf{k} \cdot \mathbf{v}_{g0} - \omega_{\mathbf{m}}) f_{\mathbf{m}} + i\hat{L}_C f_{\mathbf{m}} = \left(\left(\frac{n'_\alpha}{n_\alpha} - \frac{3T'_\alpha}{2T_\alpha} + \frac{T'_\alpha}{T_\alpha}\frac{m_\alpha v^2}{2T_\alpha}\right)\frac{ck_\perp}{B} - \frac{e_\alpha k_\parallel v_\parallel}{T_\alpha}\right) f_0 \Phi_{\mathbf{m}}, \tag{10}$$

where we denoted

$$\mathbf{k} = m\nabla\vartheta + n\nabla\varphi, \qquad k_\perp = \mathbf{k} \cdot \mathbf{h} \times \nabla r, \qquad k_\parallel = \mathbf{k} \cdot \mathbf{h}, \tag{11}$$

and ignored in the r.h.s. term with B' which enters $n'_\alpha/n_\alpha - \mu B'/T_\alpha + \dots$ for the same reason as the magnetic drift. We have also denoted the kinetic energy as

$$\frac{m_\alpha v^2}{2} = \mu B + \frac{m_\alpha v_\parallel^2}{2}.$$

We consider high frequency short scale perturbations such that $\max(\omega_{\mathbf{m}}, k_\perp V_E) \gg \nu$ where ν is the collision frequency. For these perturbations, transport is determined by the "collisionless" (Landau damping) mechanism such that it is sufficient to use Krook collision model, $\hat{L}_C = -\nu$, because small ν does not enter the result. Thus, Eq. (10) becomes algebraic,

$$\left(k_\parallel v_\parallel + k_\perp V_E - \omega_{\mathbf{m}} - i\nu\right) f_{\mathbf{m}} = \left(\left(\frac{n'_\alpha}{n_\alpha} - \frac{3T'_\alpha}{2T_\alpha} + \frac{T'_\alpha}{T_\alpha}\frac{m_\alpha v^2}{2T_\alpha}\right)\frac{ck_\perp}{B} - \frac{e_\alpha k_\parallel v_\parallel}{T_\alpha}\right) f_0 \Phi_{\mathbf{m}}, \tag{12}$$

where

$$V_E = \frac{c\Phi_0'}{B}. \tag{13}$$

Multiplying the perturbation of the distribution function (9) with radial guding center velocity $\delta\mathbf{v}_g \cdot \nabla r$ where $\delta\mathbf{v}_g$ is given by Eq. (6), averaging the result over the angles and integrating over the momentum space, we obtain particle flux density as

$$\begin{aligned}
\Gamma_\alpha &= \int \mathrm{d}^3 p f_0 \frac{1}{2} \mathrm{Re} \sum_{\mathbf{m}} \frac{ic^2 k_\perp^2 \left|\Phi_{\mathbf{m}}\right|^2}{B^2 \left(k_\parallel v_\parallel + k_\perp V_E - \omega_{\mathbf{m}} - i\nu\right)} \left(\frac{n_\alpha'}{n_\alpha} - \frac{3T_\alpha'}{2T_\alpha} + \frac{T_\alpha'}{T_\alpha}\frac{m_\alpha v^2}{2T_\alpha} - \frac{e_\alpha k_\parallel v_\parallel B}{T_\alpha c k_\perp}\right) \\
&\to -\frac{\pi}{2} \int \mathrm{d}^3 p f_0 \sum_{\mathbf{m}} \frac{c^2 k_\perp^2 \left|\Phi_{\mathbf{m}}\right|^2}{B^2} \delta\left(k_\parallel v_\parallel + k_\perp V_E - \omega_{\mathbf{m}}\right) \left(\frac{n_\alpha'}{n_\alpha} - \frac{3T_\alpha'}{2T_\alpha} + \frac{T_\alpha'}{T_\alpha}\frac{m_\alpha v^2}{2T_\alpha} - \frac{e_\alpha k_\parallel v_\parallel B}{T_\alpha c k_\perp}\right) \\
&= -\frac{\pi}{2} \int \mathrm{d}^3 p f_0 \sum_{\mathbf{m}} \frac{c^2 k_\perp^2 \left|\Phi_{\mathbf{m}}\right|^2}{B^2} \delta\left(k_\parallel v_\parallel + k_\perp V_E - \omega_{\mathbf{m}}\right) \left(A_1 + \frac{m_\alpha v^2}{2T_\alpha} A_2 - \frac{\omega_{\mathbf{m}}}{k_\perp D_B}\right),
\end{aligned} \tag{14}$$

where we introduced the following notation for the thermodynamic forces and Bohm diffusion coefficient,

$$A_1 = \frac{n_\alpha'}{n_\alpha} + \frac{e_\alpha \Phi_0'}{T_\alpha} - \frac{3T_\alpha'}{2T_\alpha}, \qquad A_2 = \frac{T_\alpha'}{T_\alpha}, \qquad D_B = \frac{cT_\alpha}{e_\alpha B}. \tag{15}$$

Integration over the momentum space,

$$\int \mathrm{d}^3 p = 2\pi m_\alpha^2 B \int\limits_0^\infty \mathrm{d}\mu \int\limits_{-\infty}^\infty \mathrm{d}v_\parallel,$$

is trivial over μ and easy over $v_\parallel$ where the δ-function should be used to give

$$\begin{aligned}
\Gamma_\alpha &= -n_\alpha \left(\frac{\pi m_\alpha}{8T_\alpha}\right)^{1/2} \int\limits_{-\infty}^\infty \mathrm{d}v_\parallel \exp\left(-\frac{m_\alpha v_\parallel^2}{2T_\alpha}\right) \sum_{\mathbf{m}} \frac{c^2 k_\perp^2 \left|\Phi_{\mathbf{m}}\right|^2}{B^2} \delta\left(k_\parallel v_\parallel + k_\perp V_E - \omega_{\mathbf{m}}\right) \\
&\times \left(A_1 + A_2 + \frac{m_\alpha v_\parallel^2}{2T_\alpha} A_2 - \frac{\omega_{\mathbf{m}}}{k_\perp D_B}\right) \\
&= -n_\alpha \left(\frac{\pi m_\alpha}{8T_\alpha}\right)^{1/2} \sum_{\mathbf{m}} \frac{c^2 k_\perp^2 \left|\Phi_{\mathbf{m}}\right|^2}{|k_\parallel| B^2} \exp\left(-\frac{m_\alpha \left(\omega_{\mathbf{m}} - k_\perp V_E\right)^2}{2k_\parallel^2 T_\alpha}\right) \\
&\times \left(A_1 + A_2 + \frac{m_\alpha \left(\omega_{\mathbf{m}} - k_\perp V_E\right)^2}{2k_\parallel^2 T_\alpha} A_2 - \frac{\omega_{\mathbf{m}}}{k_\perp D_B}\right) \\
&\equiv -n_\alpha (D_{11}^\alpha A_1 + D_{12}^\alpha A_2 - V_1^\alpha),
\end{aligned} \tag{16}$$

where the last expression defines diffusion coefficients and convection velocity. In particular, particle diffusion coeficient follows from this expression as

$$D_{11}^\alpha = \left(\frac{\pi m_\alpha}{8T_\alpha}\right)^{1/2} \sum_{\mathbf{m}} \frac{c^2 k_\perp^2 \left|\Phi_{\mathbf{m}}\right|^2}{|k_\parallel| B^2} \exp\left(-\frac{m_\alpha \left(\omega_{\mathbf{m}} - k_\perp V_E\right)^2}{2k_\parallel^2 T_\alpha}\right). \tag{17}$$

B. Static noise on the GORILLA grid

For numerical experiments in GORILLA, we generate the potential perturbation by adding in each grid node k a random potential value using the standard random numbers ξ,

$$\delta\Phi_k = \Phi_A \xi_k, \qquad \overline{\xi_k} = 0, \qquad \overline{\xi_k \xi_{k'}} = \delta_{k,k'}. \tag{18}$$

In a 1D exapmple, potential perturbation is then given by the continuous piecewise-linear periodic function of an angle $\delta\Phi(\vartheta)$. Let us compute its Fourier amplitude,

$$\Phi_m = \frac{1}{2\pi}\int\limits_0^{2\pi} \mathrm{d}\vartheta \mathrm{e}^{-im\vartheta}\delta\Phi(\vartheta) = -\frac{1}{2\pi m^2}\int\limits_0^{2\pi} \mathrm{d}\vartheta \mathrm{e}^{-im\vartheta}\delta\Phi''(\vartheta). \tag{19}$$

Second derivative $\delta\Phi''(\vartheta)$ of the continuous piecewise-linear function $\delta\Phi(\vartheta)$ equals to the sum of δ-functions at the grid nodes $\vartheta = 2\pi k/K$ where K is grid size and $1 \le k \le K$,

$$\delta\Phi''(\vartheta) = \frac{K\Phi_A}{2\pi}\sum_{k=1}^{K}\left(\xi_{k+1} + \xi_{-1} - 2\xi_k\right)\delta\left(\vartheta - \frac{2\pi k}{K}\right). \tag{20}$$

Respectively, Fourier amplitude (19) is

$$\Phi_m = -\frac{K\Phi_A}{(2\pi m)^2}\sum_{k=1}^{K}\left(\xi_{k+1} + \xi_{k-1} - 2\xi_k\right) \tag{21}$$

Obviously, its expectation value is zero, $\overline{\Phi_m} = 0$. We are interested, however, in the expectation values of its module square,

$$\overline{|\Phi_m|^2} = \frac{K^2\Phi_A^2}{(2\pi m)^4}\sum_{k,k'=1}^{K}\overline{\left(\xi_{k+1} + \xi_{k-1} - 2\xi_k\right)\left(\xi_{k'+1} + \xi_{k'-1} - 2\xi_{k'}\right)}\exp\left(\frac{2\pi i m(k-k')}{K}\right). \tag{22}$$

Using the properties (18) of random numbers we get

$$\overline{\left(\xi_{k+1} + \xi_{k-1} - 2\xi_k\right)\left(\xi_{k'+1} + \xi_{k'-1} - 2\xi_{k'}\right)} = 6\delta_{k,k'} - 4\delta_{k-1,k'} - 4\delta_{k+1,k'} + \delta_{k-2,k'} + \delta_{k+2,k'},$$

so that (22) becomes

$$\overline{|\Phi_m|^2} = \frac{K^3\Phi_A^2}{(2\pi m)^4}\left(6 - 8\cos\left(\frac{2\pi m}{K}\right) + 2\cos\left(\frac{4\pi m}{K}\right)\right). \tag{23}$$

I am lazy to compute this for a 2D grid in (ϑ, φ) variables with the size $K \times L$. Therefore I write the respectve fomula without a proof,

$$\begin{aligned}\overline{|\Phi_{\mathbf{m}}|^2} &= \frac{K^3 L^3 \Phi_A^2}{(2\pi m)^4 (2\pi n)^4}\left(6 - 8\cos\left(\frac{2\pi m}{K}\right) + 2\cos\left(\frac{4\pi m}{K}\right)\right) \\ &\times \left(6 - 8\cos\left(\frac{2\pi n}{L}\right) + 2\cos\left(\frac{4\pi n}{L}\right)\right).\end{aligned} \tag{24}$$

Let us estimate a diffusion coefficient (17) using expectation value (24) for the amplitude $|\Phi_{\mathbf{m}}|^2$ setting $\omega_{\mathbf{m}} = 0$ becuse of static perturbations and setting also $V_E = 0$. The latter setting corresponds to zero equilibrium electric field, however even for the finite but reasonable electric field this term is of the order of $V_E^2 k_\perp^2/(v_{T\alpha}^2 k_\parallel^2) \sim (k_\perp^2 \rho_{L\alpha^2})/(k_\parallel^2 r^2) \ll 1$ for $k_\parallel \sim k_\perp$. Thus we set the exponent in (17) to one. Approximating roughly $k_\parallel \approx n/R_0$ and $k_\perp \approx m/r$ we get

$$\begin{aligned} D_{11}^\alpha \approx& \left(\frac{\pi m_\alpha}{8T_\alpha}\right)^{1/2} \frac{c^2 R_0 \Phi_A^2}{r^2 B^2} \frac{1}{KL} \sum_{\mathbf{m}} \frac{m^2}{|n|} \left(\frac{2\pi m}{K}\right)^{-4} \left(\frac{2\pi n}{L}\right)^{-4} \\ &\times \left(6 - 8\cos\left(\frac{2\pi m}{K}\right) + 2\cos\left(\frac{4\pi m}{K}\right)\right)\left(6 - 8\cos\left(\frac{2\pi n}{L}\right) + 2\cos\left(\frac{4\pi n}{L}\right)\right). \end{aligned} \tag{25}$$

since $2\pi/K \ll 1$ and $2\pi/L \ll 1$, we replace summation by the integration introducing the variables $x = 2\pi m/K$ and $y = 2\pi n/L$ so that

$$\frac{1}{KL}\sum_{\mathbf{m}} \approx \frac{1}{4\pi^2}\int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y,$$

and

$$\begin{aligned} D_{11}^\alpha \approx& \left(\frac{\pi m_\alpha}{8T_\alpha}\right)^{1/2} \frac{c^2 R_0 \Phi_A^2}{r^2 B^2} \frac{K^2}{8\pi^3 L} \int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y \frac{1}{x^2 |y|^5} \\ &\times \left(6 - 8\cos(x) + 2\cos(2x)\right)\left(6 - 8\cos(y) + 2\cos(2y)\right). \end{aligned} \tag{26}$$

We note that second line of Eq. (26) scales at small x and y as $x^4 y^4$. Thus, integral over x strictly converges giving

$$\int_{-\infty}^{\infty} \mathrm{d}x \frac{6 - 8\cos(x) + 2\cos(2x)}{x^2} = \int_{-\infty}^{\infty} \mathrm{d}x \frac{8\sin(x) - 4\sin(2x)}{x} = 4\int_{-\infty}^{\infty} \mathrm{d}x \frac{\sin x}{x} = 4\pi$$

but integral over y logarithmically diverges. This divergence however is unphysical because it is the result of using collisionless limit for low $k_\parallel$ within the assumption $k_\parallel v_\parallel \gg \nu$. We can introduce some minimum value of y

$$\int_{-\infty}^{\infty} \mathrm{d}y \frac{6 - 8\cos(y) + 2\cos(2y)}{|y|^5} \to \int_{y_{\min}}^{\infty} \mathrm{d}y \frac{6 - 8\cos(y) + 2\cos(2y)}{y^5} \sim \int_{y_{\min}}^{1} \mathrm{d}y \frac{1}{y} = |\log(y_{\min})|.$$

Thus we get an estimate

$$D_{11}^\alpha \sim \frac{|\log(y_{\min})|}{2^{5/2}\pi^{3/2}} \left(\frac{m_\alpha}{T_\alpha}\right)^{1/2} \frac{c^2 R_0 \Phi_A^2}{r^2 B^2} \frac{K^2}{L}. \tag{27}$$